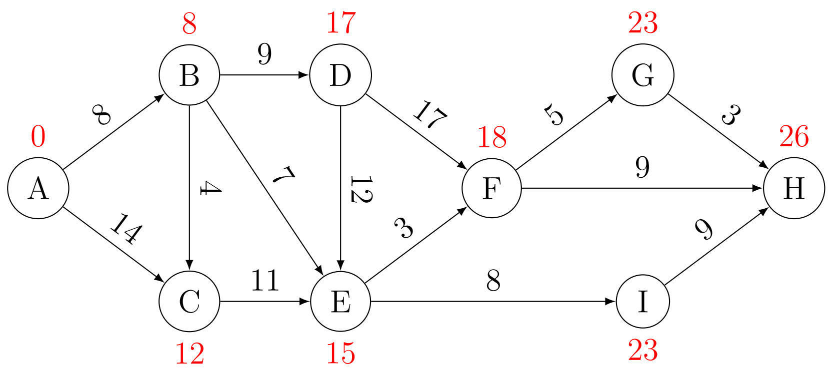
Shortest path problem

Michel Bierlaire

Solution of the practice quiz



1. The optimality conditions for a shortest path P are

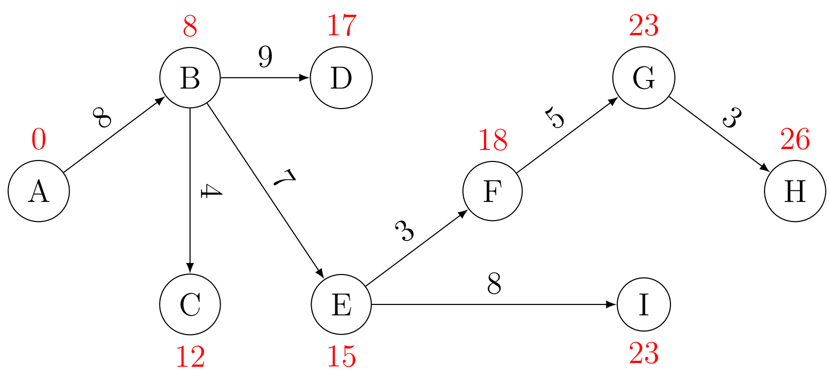
$$\lambda_j \leq \lambda_i + c_{ij}, \qquad \forall (i,j) \in \mathcal{A},$$
$$\lambda_j = \lambda_i + c_{ij}, \qquad \forall (i,j) \in P.$$

We calculate these quantities for each arc, and compare them.

(i,j)	λ_j		$\lambda_i + c_{ij}$
(A,B)	8	$=$	8,
(A,C)	12	$\leq$	14,
(B,C)	12	$=$	12,
(B,D)	17	$=$	17,
(B,E)	15	$=$	15,
(C,E)	15	$\leq$	23,
(D,E)	15	$\leq$	29,
(D,F)	18	$\leq$	34,
(E,F)	18	$=$	18,
(E,I)	23	$=$	23,
(F,G)	23	$=$	23,
(F,H)	26	$\leq$	27,
(G,H)	26	$=$	26,
(I,H)	26	$\leq$	32.

Note that the condition $\lambda_j \leq \lambda_i + c_{ij}$ is verified for all the arcs in the network.

We consider the subnetwork composed of the arcs that verify the equality condition. We obtain



This is the shortest path tree. For instance, the shortest path between A and D is the only path between A and D in this tree, that is $A \to B \to D$. Indeed, the optimality conditions are verified for this path:

- as noticed above, the condition $\lambda_j \leq \lambda_i + c_{ij}$ is verified for all the arcs in the network,

- for the two arcs of path $A \rightarrow B \rightarrow D$, the condition $\lambda_j = \lambda_i + c_{ij}$ is verified, by construction of the tree.

The shortest paths between node A and each other node are

- $A \rightarrow B$,
- $A \rightarrow B \rightarrow C$,
- $A \rightarrow B \rightarrow D$,
- $A \rightarrow B \rightarrow E$,
- $A \rightarrow B \rightarrow E \rightarrow F$,
- $A \rightarrow B \rightarrow E \rightarrow F \rightarrow G$,
- $A \rightarrow B \rightarrow E \rightarrow F \rightarrow G \rightarrow H$,
- $A \rightarrow B \rightarrow E \rightarrow I$.

2. When the set of labels verify the optimality conditions, the length of a shortest path between A and any node i is

$$\lambda_i - \lambda_A = \lambda_i,$$

as $\lambda_A = 0$. Therefore, the length of the shortest path between A and H is $\lambda_H = 26$. It can be verified by considering the path itself, obtained from the shortest path tree:

$$A \rightarrow B \rightarrow E \rightarrow F \rightarrow G \rightarrow H.$$

Its length is

$$8 + 7 + 3 + 5 + 3 = 26.$$